

Summer Training Project Report

on

“Real estate price trend analysis”

Submitted to

DEPARTMENT OF COMMERCE

In partial fulfillment of the requirement for the

award of degree of

Bachelor of Commerce- B.Com (Hons.)

Submitted by

Gurpreet kaur

2531118

Supervisor

Anut Rana

data analysis



CHANDIGARH BUSINESS SCHOOL OF ADMINISTRATION

(An Autonomous Institute affiliated to IKGPTU)

LANDRAN, MOHALI

(2025-2028)

CERTIFICATE

STUDENT DECLARATION

I, “Gurpreet kaur ”, hereby declare that I have undergone my summer training at “TCILIT ” from_(start date) to_(end date). I have completed a research project titled “real estate price trend analysis” under the guidance of Mr. Anut rana .

Further I hereby confirm that the work presented herein is genuine and original and has not been published elsewhere.

Gur _____

(Student name and Signature)

FACULTY DECLARATION

I hereby declare that the student Mr. / Ms. _____ of **Course (Semester)** has undergone his/her summer training under my periodic guidance on the Project titled “_____ (Project Title)”.

Further I hereby declare that the student was periodically in touch with me during his/her training period and the work done by student is genuine & original.

(Signature of Supervisor)

(Signature of Head of Department)

ACKNOWLEDGEMENT

I acknowledge with deep gratitude and thank the Chandigarh Business School of Administration, Landran whose course curriculum enhanced me to fulfil my **Internship** and provided me with an opportunity to learn and complete my research.

I am thankful to **Company Guide Name, Company Name** for his trust and encouragement for successful completion of the research study.

I express my sincere thanks to my research mentor **Faculty Mentor Name**, for providing me with the necessary guidance and facilities to complete my research. He /She has always been so kind to me, taught me knowledge and granted an awesome environment to my study and research. I'm so grateful for her encouraging support.

I would also like to thank all my teachers and professors whose unconditional support and guidance has helped me in completion of my research work. Last but not the least, I am thankful to my parents, classmates, and other friends for supporting me and their continuous help.

Gurpreet kaur

TABLE OF CONTENTS

Certificate by Company Guide		ii
Student Declaration		iii
Faculty Declaration		iv
Acknowledgment		v
CHAPTER NO.	CHAPTER TITLE	PAGE NO.
1	Introduction to the Company	
2	Introduction to the Research Problem	
3	Need, Scope and Objectives of the Study	
4	Research Methodology	
5	Data Analysis and Interpretation	
6	Findings of the Study	
7	Conclusion, Suggestions & Recommendations of the Study	
References and Bibliography		
Appendix (Questionnaire, Glossary of Terms, Abbreviations, Documents, Performa, Financial statements, etc.)		

LIST OF TABLES

TABLE NO.	TABLE TITLE	PAGE NO.

LIST OF FIGURES

FIGURE NO.	FIGURE TITLE	PAGE NO.

INTRODUCTION

Real Estate Price Trend Analysis is the process of examining historical and current property prices to identify patterns, trends, and factors that influence the value of real estate over time. It helps buyers, sellers, investors, developers, and policymakers understand how property prices change across different locations and periods, enabling informed decision-making.

The real estate market is influenced by a wide range of factors, including economic conditions, population growth, interest rates, infrastructure development, government policies, employment opportunities, and local demand and supply. By analyzing these factors alongside historical price data, it is possible to forecast future property values and identify profitable investment opportunities.

With the advancement of data analytics and machine learning, real estate price trend analysis has become more accurate and efficient. Techniques such as statistical analysis, regression models, time-series forecasting, and predictive machine learning algorithms are widely used to uncover hidden patterns in property data and estimate future market trends.

The primary objectives of real estate price trend analysis include:

- Understanding historical property price movements.
- Identifying factors that significantly affect property values.
- Predicting future real estate prices.
- Supporting investment and purchasing decisions.
- Assisting government agencies and urban planners in policy formulation.

Overall, real estate price trend analysis plays a crucial role in reducing investment risk, improving market transparency, and helping stakeholders make data-driven decisions in the dynamic real estate market.

INTRODUCTION TO RESEARCH PROBLEM

Auto recovery contains some recovered files that haven't been opened.

View recovered files X

Real estate price trend analysis

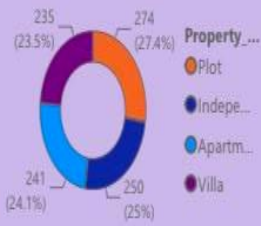
Property_Type

Apartment

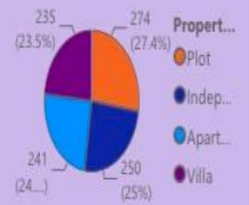
Independent House

Plot

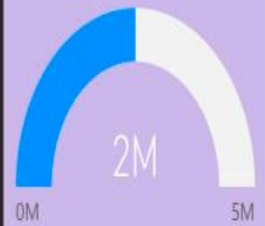
property type distribution



Count of Property_ID by Property_Type



Sum of Area_sqft



Total price

20.54bn

Total property

1K

Total Bedrooms

3K

Price per sqft

9.06M

Sum of Price_INR by City



City and Price_INR



Need for the Study

The real estate sector is one of the most significant contributors to economic growth and investment. Property prices fluctuate due to various factors such as location, economic conditions, infrastructure development, interest rates, government policies, and market demand. These fluctuations make it difficult for buyers, sellers, investors, and developers to make informed decisions based solely on intuition or limited market knowledge.

The need for this study arises from the increasing complexity of the real estate market and the growing availability of large volumes of property-related data. By analyzing historical price trends and identifying the key factors influencing property values, this study aims to provide meaningful insights into market behavior and future price movements.

Furthermore, advances in data analytics and machine learning have created opportunities to develop predictive models that can estimate future property prices with greater accuracy. Such analysis helps reduce investment risks, improves transparency in the market, and supports strategic planning for individuals, businesses, financial institutions, and government agencies.

This study is essential because it:

- Helps buyers and investors make informed property investment decisions.
- Identifies the major factors affecting real estate prices.
- Analyzes historical price trends to understand market behavior.
- Assists developers in planning projects based on market demand.
- Supports financial institutions in property valuation and loan assessment.
- Enables policymakers and urban planners to understand housing market dynamics.
- Provides a foundation for developing predictive models for future property price forecasting.

Overall, the study contributes to better decision-making by transforming real estate data into actionable insights, making the property market more efficient, transparent, and reliable.

SCOPE OF THE PROJECT

The scope of this project is to analyze historical and current real estate data to identify price trends and the factors that influence property values. The project focuses on collecting, processing, and analyzing property-related data to provide meaningful insights and support data-driven decision-making.

The scope of the project includes:

- Collecting real estate data such as property prices, location, property type, size, amenities, and transaction history.
- Analyzing historical price trends to understand market behavior over different periods.
- Identifying the key factors that affect property prices, including location, infrastructure, demand and supply, economic conditions, and government policies.
- Applying statistical and data analysis techniques to discover patterns and relationships within the data.
- Developing predictive models using machine learning or regression techniques to estimate future property prices.
- Visualizing price trends through graphs, charts, and dashboards for better understanding and interpretation.
- Comparing property prices across different regions or localities to identify high-growth areas.
- Providing insights that assist buyers, sellers, investors, developers, and policymakers in making informed decisions.

Limitations of the Scope

- The analysis depends on the availability and quality of real estate data.
- Predictions may be affected by sudden economic changes, policy updates, or unforeseen market conditions.
- The project focuses on selected regions or datasets and may not represent the entire real estate market.

Overall, this project aims to provide a comprehensive understanding of real estate price trends and support accurate forecasting using data analytics techniques, thereby helping stakeholders make better investment and planning decisions.

OBJECTIVES OF THE PROJECT

The main objective of this project is to analyze historical real estate data, identify price trends, and predict future property prices using data analysis and machine learning techniques. The project aims to provide accurate and reliable insights that support informed decision-making in the real estate market.

The specific objectives of the project are:

- To collect and organize real estate data from reliable sources.
- To analyze historical property price trends across different locations and time periods.
- To identify the key factors influencing real estate prices, such as location, property size, amenities, infrastructure, demand and supply, and economic conditions.
- To perform exploratory data analysis (EDA) to discover patterns, correlations, and market trends.
- To develop predictive models using statistical or machine learning algorithms for property price forecasting.
- To evaluate the performance of the prediction models using appropriate accuracy metrics.
- To visualize real estate trends through graphs, charts, and dashboards for easy interpretation.
- To provide useful insights that help buyers, sellers, investors, developers, and policymakers make informed decisions.
- To support efficient real estate planning by reducing uncertainty in property price estimation.

Overall, the project seeks to improve the accuracy of real estate price prediction and enhance decision-making by leveraging data analytics and predictive modeling techniques.

RESEARCH METHODOLOGY

The research methodology for the **Real Estate Price Trend Analysis** project follows a systematic approach to collect, process, analyze, and interpret real estate data for identifying price trends and predicting future property values.

1. Research Design

This study is based on a **quantitative research design**, as it involves the collection and analysis of numerical data related to real estate prices. Statistical and machine learning techniques are used to identify trends and make predictions.

2. Data Collection

The study uses **secondary data** collected from reliable sources such as:

- Real estate websites and property listing platforms.
- Government housing and land records.
- Publicly available real estate datasets.
- Research publications and market reports.

The dataset includes variables such as:

- Property price
- Location
- Property type
- Area (square feet)
- Number of bedrooms and bathrooms
- Amenities
- Year of construction
- Transaction date

3. Data Preprocessing

Before analysis, the collected data is cleaned and prepared by:

- Removing duplicate records.
- Handling missing values.
- Correcting inconsistent data.
- Encoding categorical variables.
- Normalizing or scaling numerical features where required.

4. Exploratory Data Analysis (EDA)

EDA is performed to understand the dataset and identify patterns using:

- Descriptive statistics
- Correlation analysis
- Histograms and box plots
- Scatter plots
- Heat maps
- Trend analysis over time

5. Data Analysis and Modeling

The processed data is analyzed using statistical and machine learning techniques such as:

- Linear Regression
- Multiple Regression
- Decision Tree Regression
- Random Forest Regression
- Time Series Analysis (if historical time-based data is available)

These models are used to predict future real estate prices and identify the factors that significantly influence property values.

6. Model Evaluation

The performance of the prediction models is evaluated using metrics such as:

- Mean Absolute Error (MAE)
- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)
- R-squared (R^2) Score

The model with the best performance is selected for price prediction.

7. Interpretation and Reporting

The final step involves interpreting the results, presenting visualizations, and drawing conclusions. The findings are used to provide recommendations for buyers, sellers, investors, developers, and policymakers.

Conclusion

The research methodology combines data collection, preprocessing, exploratory analysis, predictive modeling, and performance evaluation to analyze real estate price trends effectively. This structured approach ensures reliable results and supports informed decision-making in the real estate market.

DATA AND INTERPRETATION ANALYSIS

Data Analysis

Data analysis is the process of examining, organizing, and interpreting collected real estate data to identify meaningful patterns and trends. In this project, historical property data is analyzed to understand the factors influencing real estate prices and to predict future price movements.

The analysis is carried out in the following stages:

1. **Data Cleaning:** Missing values, duplicate records, and inconsistent data are identified and removed to improve data quality.
2. **Exploratory Data Analysis (EDA):** Statistical summaries and visualizations such as histograms, bar charts, scatter plots, line graphs, and heat maps are used to understand the distribution of data and identify relationships among variables.
3. **Feature Analysis:** Important features such as property location, area (square feet), number of bedrooms, number of bathrooms, property type, amenities, and transaction date are analyzed to determine their impact on property prices.
4. **Trend Analysis:** Historical price data is examined over different time periods to identify increasing, decreasing, or stable price trends in the real estate market.
5. **Predictive Analysis:** Machine learning algorithms such as Linear Regression, Decision Tree Regression, or Random Forest Regression are applied to predict future property prices based on historical data.
6. **Model Evaluation:** The prediction model is evaluated using performance metrics such as Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R-squared (R^2) to measure prediction accuracy.

Interpretation

The analysis of the real estate dataset provides valuable insights into the factors that influence property prices and market behavior.

- **Location** is one of the most significant factors affecting property prices. Properties located in well-developed areas with better infrastructure, transportation, schools, and hospitals generally have higher prices.
- **Property Size** shows a positive relationship with price. Larger properties tend to have higher market values than smaller ones.
- **Amenities** such as parking, security, parks, swimming pools, and shopping facilities increase the attractiveness and value of a property.
- **Market Trends** indicate that property prices generally increase over time due to inflation, urbanization, population growth, and increasing demand, although short-term fluctuations may occur because of economic conditions or government policies.
- **Predictive Models** can estimate future property prices with reasonable accuracy when trained on high-quality historical data, helping stakeholders make informed investment decisions.

Overall Interpretation

The study demonstrates that real estate prices are influenced by multiple economic, social, and geographical factors. By analyzing historical trends and applying predictive techniques, the project provides reliable insights into future price movements. These findings can help buyers, sellers, investors, developers, financial institutions, and policymakers make informed decisions while reducing uncertainty in the real estate market.

FINDINGS OF THE STUDY

The following are the major findings of the **Real Estate Price Trend Analysis** project:

1. **Location is the most influential factor** affecting property prices. Properties situated in prime locations with good infrastructure, transportation, educational institutions, and healthcare facilities generally have higher values.
2. **Property size has a positive impact on price.** Larger properties with greater built-up area and land area tend to command higher market prices.
3. **Amenities increase property value.** Features such as parking facilities, security, parks, swimming pools, elevators, and recreational spaces contribute to higher property prices.
4. **Real estate prices generally show an upward trend** over time due to factors such as urbanization, population growth, inflation, and increasing demand for housing.
5. **Economic conditions significantly influence the market.** Interest rates, employment levels, inflation, and government housing policies directly affect property demand and pricing.
6. **Data analysis reveals strong relationships** between property price and key variables such as location, area, number of bedrooms, and available amenities.
7. **Machine learning models can predict property prices with reasonable accuracy** when trained on high-quality historical data, making them useful for price forecasting and investment planning.
8. **Data visualization techniques** such as charts, graphs, and heat maps make it easier to identify market trends, regional price differences, and price fluctuations over time.
9. **The study supports informed decision-making** by providing valuable insights for buyers, sellers, investors, developers, financial institutions, and policymakers.
10. **Accurate and well-maintained data improves prediction performance**, highlighting the importance of data quality in real estate price analysis.

Overall Finding

The study concludes that real estate prices are influenced by a combination of location, property characteristics, economic conditions, and market demand. Analyzing historical data and applying predictive analytics enables more accurate price forecasting, helping stakeholders make better investment, planning, and purchasing decisions while reducing uncertainty in the real estate market.

CONCLUSION

The **Real Estate Price Trend Analysis** project demonstrates that property prices are influenced by a combination of factors, including location, property size, amenities, infrastructure development, market demand, economic conditions, and government policies. By analyzing historical real estate data, the study identifies significant trends and patterns that help explain fluctuations in property prices over time.

The application of data analysis and machine learning techniques enables accurate prediction of future property prices, providing valuable insights for buyers, sellers, investors, developers, financial institutions, and policymakers. The study also highlights the importance of high-quality data and proper preprocessing in improving the accuracy and reliability of predictive models.

Overall, the project shows that real estate price trend analysis is an effective tool for understanding market behavior, reducing investment risks, and supporting informed decision-making. As technology and data availability continue to improve, advanced analytical techniques can further enhance the accuracy of property price forecasting and contribute to the development of a more transparent and efficient real estate market.

SUGGESTIONS AND RECOMMENDATIONS OF THE STUDY

Based on the findings of the **Real Estate Price Trend Analysis**, the following suggestions and recommendations are proposed:

1. **Use Data-Driven Decision-Making:** Buyers, sellers, and investors should rely on data analysis and market trends rather than assumptions when making real estate decisions.
2. **Focus on Location:** Since location has a significant impact on property prices, investors should prioritize areas with good infrastructure, transportation, educational institutions, healthcare facilities, and future development potential.
3. **Regularly Update Market Data:** Real estate databases should be updated frequently to ensure that price predictions reflect current market conditions and remain accurate.
4. **Adopt Advanced Predictive Models:** Real estate companies and financial institutions should use machine learning and artificial intelligence techniques to improve the accuracy of property price forecasting.
5. **Improve Data Quality:** Accurate, complete, and reliable property data should be maintained to enhance the performance of analytical models and reduce prediction errors.
6. **Consider Economic Factors:** Investors and developers should monitor factors such as interest rates, inflation, employment levels, and government policies, as these have a direct impact on real estate prices.
7. **Promote Market Transparency:** Government agencies and regulatory authorities should encourage transparent property transactions and maintain accessible real estate records to improve market efficiency.
8. **Support Sustainable Urban Development:** Urban planners should focus on infrastructure development, affordable housing, and public amenities to promote balanced growth and stabilize property values.
9. **Use Visual Analytics:** Real estate organizations should implement dashboards and data visualization tools to monitor market trends and support faster, evidence-based decision-making.
10. **Encourage Further Research:** Future studies can include larger datasets, additional variables such as environmental and demographic factors, and advanced deep learning models to improve prediction accuracy.

Overall Recommendation

The study recommends adopting modern data analytics and machine learning techniques for real estate price analysis to improve forecasting accuracy, minimize investment risks, and support informed decision-making. Continuous monitoring of market trends, maintaining high-quality data, and considering economic and regional factors will help stakeholders make better investment, planning, and policy decisions in the real estate sector.

REFERENCES AND BIBLIOGRAPHY

The following references can be included for the **Real Estate Price Trend Analysis** project:

References

1. Data Science for Business — Provost, F., & Fawcett, T. (2013). *Data Science for Business: What You Need to Know About Data Mining and Data-Analytic Thinking*. O'Reilly Media.
2. Python for Data Analysis — McKinney, W. (2022). *Python for Data Analysis: Data Wrangling with pandas, NumPy, and Jupyter*. O'Reilly Media.
3. Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow — Géron, A. (2022). *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow*. O'Reilly Media.
4. An Introduction to Statistical Learning — James, G., Witten, D., Hastie, T., & Tibshirani, R. (2021). *An Introduction to Statistical Learning*. Springer.
5. Machine Learning — Mitchell, T. M. (1997). *Machine Learning*. McGraw-Hill.
6. World Bank — Reports and datasets related to economic indicators, housing markets, and urban development.
7. National Housing Bank — Housing market reports and real estate data resources.
8. Reserve Bank of India — Reports related to interest rates, inflation, and economic factors affecting real estate markets.

Bibliography

- Provost, F., & Fawcett, T. (2013). *Data Science for Business*. O'Reilly Media.
- McKinney, W. (2022). *Python for Data Analysis*. O'Reilly Media.
- Géron, A. (2022). *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow*. O'Reilly Media.
- James, G., Witten, D., Hastie, T., & Tibshirani, R. (2021). *An Introduction to Statistical Learning*. Springer.
- Mitchell, T. M. (1997). *Machine Learning*. McGraw-Hill.
- Research articles and journals related to real estate price prediction, data analytics, and machine learning.
- Government publications and publicly available real estate market datasets.

